Alpha Release

Project Overview

This project analyzes Max Verstappen's 2023 Formula 1 season to understand the scale and nature of his dominance. While the Red Bull car is widely considered the fastest on the grid, this project investigates whether performance differences can be explained by the car alone, or whether driver-level consistency and execution played a decisive role.

To explore this, the project combines multiple visualization approaches. It compares Verstappen directly with his teammate Sergio Pérez—who drove the same car—while also placing Verstappen's season in a broader historical and cross-sport context. The goal is to move beyond simple point totals and instead reveal how dominance unfolds across multiple performance dimensions.

Project Website

https://varun1421.github.io/Verstappen/

Completed Work (Alpha Stage)

At the Alpha stage, the project focuses on establishing the core visual and analytical framework through four key visualizations.

1. Small Multiples: Teammate Comparison (D3)

The primary visualization is a 2×6 grid of small multiples comparing Max Verstappen and Sergio Pérez across six performance metrics:

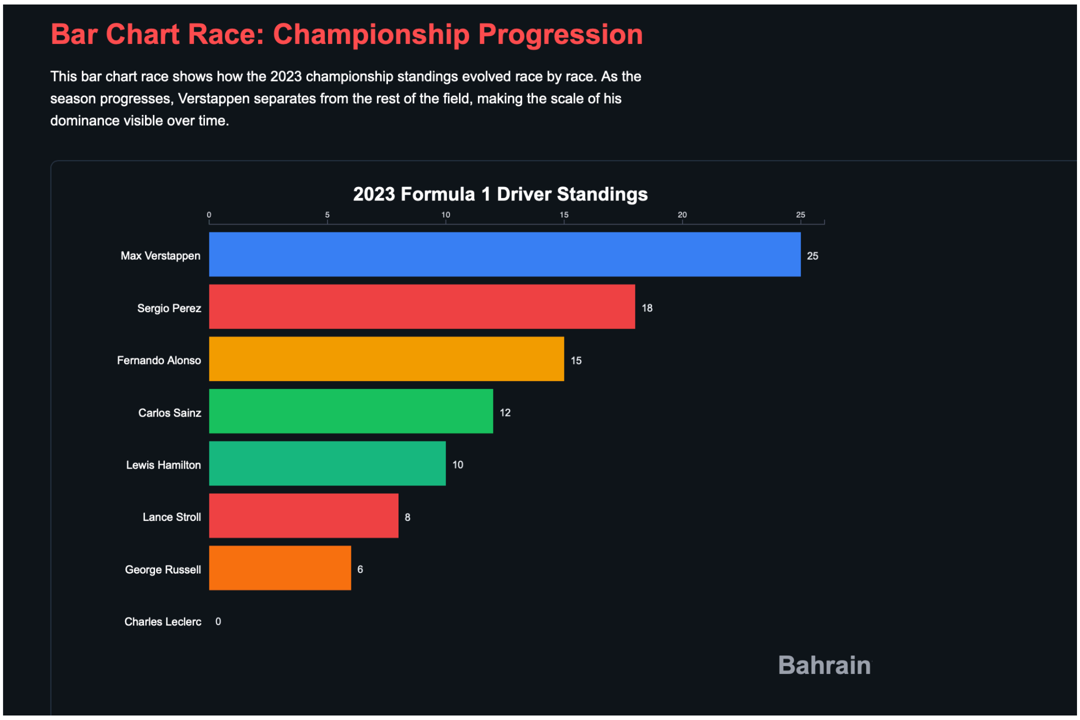
- Cumulative Points
- Cumulative Wins

- Cumulative Podiums
- Cumulative Pole Positions
- Cumulative Fastest Laps
- Points per Race

Each row represents a driver, while each column represents a metric. This layout allows for direct comparison across identical scales, isolating the impact of the driver from the car. By comparing two drivers in the same machinery, the visualization highlights differences in consistency, race execution, and the ability to convert opportunities into strong results.

This visualization is implemented using D3 and serves as the analytical backbone of the project.

2. Bar Chart Race: Championship Progression (D3)

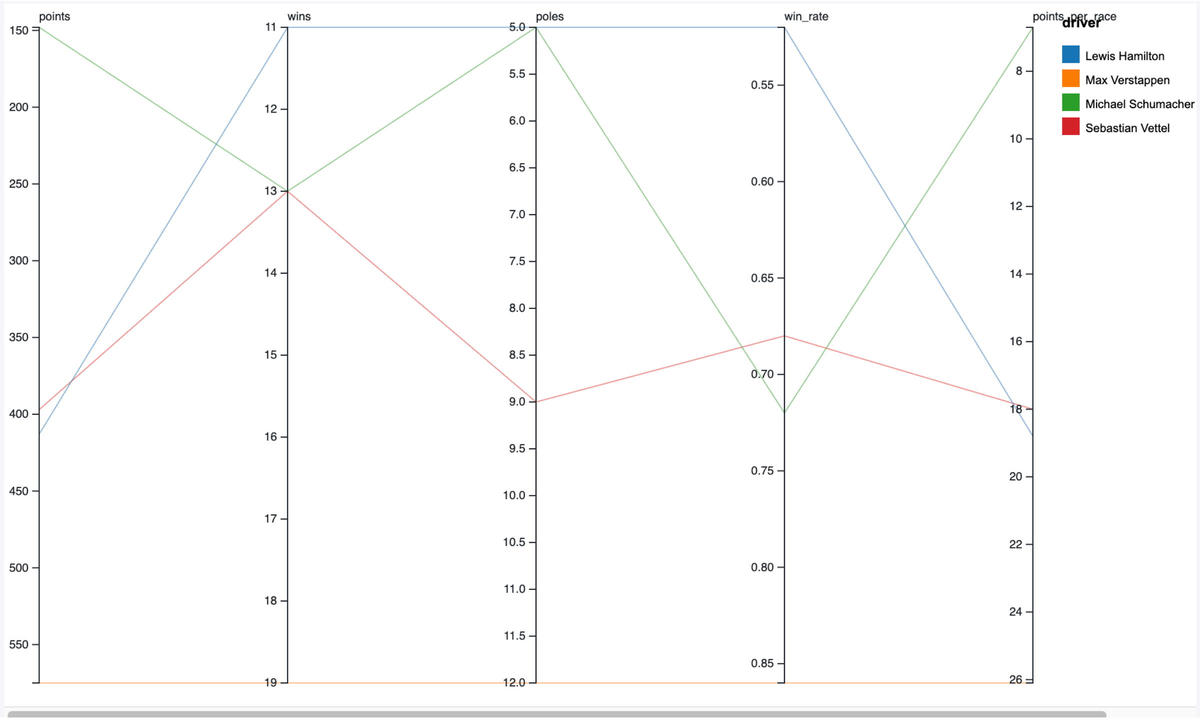


A bar chart race visualizes how cumulative championship points evolved over the 2023 season. Each frame represents a race, and drivers are ranked by total points at that moment in time.

This visualization emphasizes how Verstappen gradually separates from the rest of the field. Early races show a relatively competitive standings table, while later races reveal a clear and widening gap, making the progression of dominance visually explicit.

The chart is implemented using D3 with animated transitions between race rounds.

3. Parallel Coordinates: Historical Season Comparison (Prototype)



A parallel coordinates chart compares Verstappen’s 2023 season with other historically dominant Formula 1 seasons, including:

- Sebastian Vettel (2013)
- Michael Schumacher (2004)
- Lewis Hamilton (2019)

Each axis represents a performance metric such as wins, podiums, poles, and points per race. This allows multiple dimensions of performance to be viewed simultaneously.

At the Alpha stage, this visualization is implemented as a static prototype using RAWGraphs. It demonstrates the intended design and analytical direction for comparing multi-dimensional performance across seasons.

4. ECDF: Cross-Sport Dominance Comparison (Prototype)

An ECDF (Empirical Cumulative Distribution Function) visualization compares Verstappen’s 2023 season with dominant seasons from other sports, such as:

- Michael Jordan (NBA)
- Roger Federer (Tennis)
- Cristiano Ronaldo (Football)
- Virat Kohli? (Cricket)

Each curve will represent the distribution of normalized dominance scores within a season. Curves that lie further to the right indicate greater overall dominance.

This visualization currently hasn't been implemented as a static prototype yet as neither RAWGraphs nor DataWrapper support ECDF format.

Upcoming Milestones (Beta and Final)

The next stages of the project will focus on expanding both the depth and interactivity of the visualizations:

- Convert parallel coordinates into an interactive D3 visualization
- Create the ECDF using more rigorously normalized cross-sport metrics
- Add interactivity such as filtering, highlighting, and tooltips across all charts
- Improve visual consistency and design across the entire site
- Integrate all visualizations into a cohesive narrative flow

Challenges and Roadblocks

The primary challenge during the Alpha stage was data preparation. While race results are publicly available, the data required significant restructuring to support multiple visualization types. In particular, constructing cumulative metrics and aligning race-by-race data across drivers required careful validation.

Another challenge was balancing scope and implementation. The project involves multiple complex visualizations, and it was necessary to prioritize building a strong foundation (small multiples and bar chart race) while representing more advanced ideas (parallel coordinates and ECDF) as prototypes.

Summary

The Alpha release establishes the core analytical direction of the project. It demonstrates that Verstappen's dominance cannot be fully explained by the car alone, using direct teammate comparison and temporal progression. It also lays the groundwork for broader comparisons across seasons and sports, which will be developed further in subsequent stages.